3.3 Exercises

1–22 Differentiate.

1. $f(x) = 3\sin x - 2\cos x$
2. $f(x) = \tan x - 4\sin x$
3. $y = x^2 + \cot x$
4. $y = 2\sec x - \csc x$
5. $h(\theta) = \theta^2 \sin\theta$
6. $g(x) = 3x + x^2\cos x$
7. $y = \sec\theta\,\tan\theta$
8. $y = \sin\theta\,\cos\theta$
9. $f(\theta) = (\theta - \cos\theta)\sin\theta$
10. $g(\theta) = e^\theta(\tan\theta - \theta)$
11. $H(t) = \cos^2 t$
12. $f(x) = e^x\sin x + \cos x$
13. $f(\theta) = \dfrac{\sin\theta}{1+\cos\theta}$
14. $y = \dfrac{\cos x}{1-\sin x}$
15. $y = \dfrac{x}{2-\tan x}$
16. $f(t) = \dfrac{\cot t}{e^t}$
17. $f(w) = \dfrac{1+\sec w}{1-\sec w}$
18. $y = \dfrac{\sin t}{1+\tan t}$
19. $y = \dfrac{t\sin t}{1+t}$
20. $g(z) = \dfrac{z}{\sec z + \tan z}$
21. $f(\theta) = \theta\cos\theta\,\sin\theta$
22. $f(t) = te^t\cot t$

27–30 Find an equation of the tangent line to the curve at the given point.

27. $y = \sin x + \cos x, \quad (0, 1)$

28. $y = x + \sin x, \quad (\pi, \pi)$

29. $y = e^x \cos x + \sin x, \quad (0, 1)$

30. $y = \dfrac{1 + \sin x}{\cos x}, \quad (\pi, -1)$

3.4 Exercises

1–6 Write the composite function in the form $f(g(x))$. [Identify the inner function $u = g(x)$ and the outer function $y = f(u)$.] Then find the derivative dy/dx.

1. $y = (5 - x^4)^3$

2. $y = \sqrt{x^3 + 2}$

3. $y = \sin(\cos x)$

4. $y = \tan(x^2)$

5. $y = e^{\sqrt{x}}$

6. $y = \sqrt[3]{e^x + 1}$

Worksh

7–52 Find the derivative of the function.

7–17

7. $f(x) = (2x^3 - 5x^2 + 4)^5$

8. $f(x) = (x^5 + 3x^2 - x)^{50}$

9. $f(x) = \sqrt{5x + 1}$

10. $f(x) = \dfrac{1}{\sqrt[3]{x^2 - 1}}$

11. $g(t) = \dfrac{1}{(2t + 1)^2}$

12. $F(t) = \left(\dfrac{1}{2t + 1}\right)^4$

13. $f(\theta) = \cos(\theta^2)$

14. $g(\theta) = \cos^2 \theta$

15. $g(x) = e^{x^2 - x}$

16. $y = 5^{\sqrt{x}}$

57–60 Find an equation of the tangent line to the curve at the given point.

57. $y = 2^x$; $(0, 1)$

58. $y = \sqrt{1 + x^3}$, $(2, 3)$

59. $y = \sin(\sin x)$, $(\pi, 0)$

60. $y = xe^{-x^2}$, $(0, 0)$

5–22 Find dy/dx by implicit differentiation.

5. $x^2 - 4xy + y^2 = 4$

6. $2x^2 + xy - y^2 = 2$

7. $x^4 + x^2y^2 + y^3 = 5$

8. $x^3 - xy^2 + y^3 = 1$

2–26 Differentiate the function.

Worksheet

2. $g(t) = \ln(3 + t^2)$

3. $f(x) = \ln(x^2 + 3x + 5)$

4. $f(x) = x \ln x - x$

5. $f(x) = \sin(\ln x)$

6. $f(x) = \ln(\sin^2 x)$

7. $f(x) = \ln \dfrac{1}{x}$

8. $y = \dfrac{1}{\ln x}$

9. $g(x) = \ln(xe^{-2x})$

10. $g(t) = \sqrt{1 + \ln t}$

11. $F(t) = (\ln t)^2 \sin t$

12. $p(t) = \ln \sqrt{t^2 + 1}$

13. $y = \log_8(x^2 + 3x)$

14. $y = \log_{10} \sec x$

15. $F(s) = \ln \ln s$

16. $P(v) = \dfrac{\ln v}{1 - v}$

63–78 Find the derivative of the function. Simplify where possible.

63. $f(x) = \sin^{-1}(5x)$

64. $g(x) = \sec^{-1}(e^x)$

65. $y = \tan^{-1}\sqrt{x - 1}$

66. $y = \tan^{-1}(x^2)$

67. $y = (\tan^{-1}x)^2$

68. $g(x) = \arccos\sqrt{x}$